

Coupled rigid contact and prescribed moving containers

Physics-engine research note

4 October 2026

This note defines the mechanics and verification contract used by the moving-container experiments. The algebra is established mechanics, not a novel collision theory. Numerical tests do not establish measured material accuracy. The independent normal solve below is implemented; a complete global frictional, adaptive error-controlled engine is future work.

1 A consistent wedge in two and three dimensions

Positive contact action goes from body B to body A. Each lever arm is measured from its body's centre of mass to the same world contact location. Define

$$r^\wedge = \begin{bmatrix} -r_y & r_x \end{bmatrix} \quad \text{in 2D,} \quad (1)$$

$$r^\wedge = \begin{bmatrix} 0 & -r_z & r_y \\ r_z & 0 & -r_x \\ -r_y & r_x & 0 \end{bmatrix} \quad \text{in 3D.} \quad (2)$$

Then angular impulse from the force impulse is $r^\wedge \Delta p$, and point velocity is $v + (r^\wedge)^\top \omega$ in both dimensions. In 2D spin and angular impulse are scalars. In 3D they have three components; the inertia tensor $\mathbb{I}$ is expressed in world coordinates. With a free couple impulse ΔL at the contact,

$$\Delta p_A = \Delta p, \quad \Delta p_B = -\Delta p, \quad \Delta L_A = \Delta L + r_A^\wedge \Delta p, \quad \Delta L_B = -\Delta L - r_B^\wedge \Delta p. \quad (3)$$

An ideal frictional point carries force but no free couple. A nonzero free couple represents a finite contact patch or an explicitly declared resistance law.

2 Point mobility and kinetic energy

For body i , the point velocity/spin response is

$$\begin{pmatrix} \Delta v_{c,i} \\ \Delta \omega_i \end{pmatrix} = M_i \begin{pmatrix} \Delta p_i \\ \Delta L_{c,i} \end{pmatrix}, \quad M_i = \begin{pmatrix} m_i^{-1} + (r_i^\wedge)^\top \mathbb{I}_i^{-1} r_i^\wedge & (r_i^\wedge)^\top \mathbb{I}_i^{-1} \\ \mathbb{I}_i^{-1} r_i^\wedge & \mathbb{I}_i^{-1} \end{pmatrix}. \quad (4)$$

Scalar additions to square matrix blocks imply the appropriate identity. For two freely responding bodies, $M = M_A + M_B$. Set

$$V = \begin{pmatrix} v_{c,A} - v_{c,B} \\ \omega_A - \omega_B \end{pmatrix}, \quad \Delta P = \begin{pmatrix} \Delta p \\ \Delta L \end{pmatrix}. \quad \Delta V = M \Delta P, \quad \Delta T = (V^-)^\top \Delta P + \frac{1}{2} \Delta P^\top M \Delta P. \quad (5)$$

For a reference length $\ell > 0$, use $\tilde{V} = (v_c, \ell \omega)$ and $\tilde{\Delta P} = (\Delta p, \Delta L/\ell)$. This preserves their work pairing. Consistent inertia and mobility scaling makes the solution independent of ℓ ; this length is not a material rolling radius.

3 Many contacts and external work

Let U collect scaled dynamic centre velocities/spins and H the block body mass matrix. A contact incidence/Jacobian G maps body motion to relative point motion. Each contact point has its own row block, even when several points join the same bodies. Separate prescribed boundary motion:

$$V^- = GU^- + V_{\text{prescribed}}, \quad K = GH^{-1}G^\top, \quad (6)$$

$$U^+ = U^- + H^{-1}G^\top \Delta P, \quad V^+ = V^- + K \Delta P. \quad (7)$$

These are sparse body/contact operators. Shared bodies produce off-diagonal coupling; zero entries encode disconnected contacts. K may be singular: redundant contacts can have nonunique pressure impulses and a unique body response. Never require an inverse of the full contact matrix. With frozen geometry and no other impulsive action,

$$W_{\text{boundary}} = -V_{\text{prescribed}}^\top \Delta P, \quad (8)$$

$$\Delta T_{\text{dynamic}} = W_{\text{boundary}} + (V^-)^\top \Delta P + \frac{1}{2} \Delta P^\top K \Delta P. \quad (9)$$

Therefore passive contact bounds the last two terms by zero, not necessarily the contents' kinetic-energy change. For a box translating at constant b with zero gravity, the finite-history identity is $W = b \cdot \Delta p_{\text{contents}}$. Changing or rotating boundary motion requires a time-resolved reaction/work ledger. State histories alone do not supply that exact ledger.

4 Implemented normal oracle and friction limits

For fixed frictionless contacts, zero restitution and no other forces, the implemented normal oracle solves

$$\min_{j_n \geq 0} \frac{1}{2} j_n^\top K_{nn} j_n + (v_n^-)^\top j_n, \quad v_n^+ = v_n^- + K_{nn} j_n \geq 0, \quad j_n \odot v_n^+ = 0. \quad (10)$$

Residual tests gate acceptance after optimisation and active-set correction. This is a frozen, dense verification solve, not a measured scalable engine. For one isolated uncoupled normal contact,

$$j_n = -\frac{(1 + e_n)v_n^-}{K_{nn}}, \quad 0 \leq e_n \leq 1, \text{quad} v_n^- < 0. \quad (11)$$

With tangent or spin coupling, this scalar formula alone is insufficient. With many contacts, independently prescribed restitution targets can be incompatible or energetic; bounded coefficients do not prevent that.

A Coulomb law permits sticking with $v_t = 0$ and $\|f_t\| \leq \mu_s f_n$. During sliding, $f_t = -\mu_d f_n v_t / \|v_t\|$, with $0 \leq \mu_d \leq \mu_s$. Static capacity permits sticking; it does not supply elastic reversal. A positive tangential restitution target is rebound, not a final sticking state. Elastic rebound needs stored tangential contact energy/history or an explicitly calibrated impact approximation. Analogous patch rotation needs its own stored state and dissipative torque, with a dimensional capacity such as $\|\tau_r\| \leq \rho_r f_n$, where ρ_r has units of length. In 3D rolling and normal-axis twisting are distinct channels; a point has no normal angular-friction channel. In 2D there is one out-of-plane couple.

The native comparators use their established single friction coefficient, geometric-mean mixing and maximum restitution mixing. This is not the proposed separate static/dynamic/history law. Their numerical softness is not a measured Young modulus. Coefficient fitting must not hide missed collisions or insufficient constraint convergence. Freeze physical parameters across fidelity comparisons.

5 Exact packed-row test and adaptation contract

Put N touching unit-mass disks in a closed box moving at $b = (1, 0)$ while the disks initially rest. With all contacts present, no deformation and zero restitution, the left wall, chain and right wall give

$$1 \leq v_{1x}^+ \leq v_{2x}^+ \leq \dots \leq v_{Nx}^+ \leq 1. \quad \Delta p_x = N, \quad \Delta T = N/2, \quad W = N, \quad D = N/2. \quad (12)$$

Thus every disk, not only total momentum, is independently checkable. Native first-frame errors include contact discovery and floating geometry. Real elastic materials instead transmit finite-speed waves; this test verifies a declared rigid simultaneous-constraint model, not acoustic material dynamics.

Adapt numerical work within one physical model first: retain state, contact caches and elastic history, and vary collision updates/solver iterations. Do not treat changes in friction, stiffness or restitution as accuracy switches. Use per-island unilateral/friction residuals, geometry checks, energy/work checks and refinement/shadow comparisons. Residuals alone do not bound future trajectory error. Discontinuous impacts need observable tolerances and event timing tests; smooth-region integration order is a separate claim. The current adaptive controller remains a heuristic. The measured study does not establish automatic error-controlled fidelity selection.

6 Evidence and next method choice

The companion report retains every trace and separate refinement gates. Box2D provides a reproducible fast baseline; coupled sparse contact solves are the next investigation for large rigid islands, with compliant contact as a separately calibrated physical model when wave propagation and rebound history matter. There is no universal most accurate/performance-optimal contact method. Partitioning a single rigid body only labels material patches. Deformation requires independent cell/internal degrees of freedom and objective compliant interfaces, validated against static compliance, modes and held-out impacts.

References

- [1] E. Catto, *Solver2D* (2024). <https://box2d.org/posts/2024/02/solver2d/>.
- [2] Box2D simulation documentation. https://box2d.org/documentation/md_simulation.html.
- [3] MuJoCo computation documentation, constraints and contact model. <https://mujoco.readthedocs.io/en/stable/computation/index.html>.
- [4] Critical and supportive prior-art assessments and joint verdict, including Stronge frictional-impact energetics and Maw–Barber–Fawcett tangential compliance. References and checked metadata are retained in `research/critical-review.md` and `publication-case.md`.